Source: docs\technical\delivery-architecture.md | Export: 11 November 2026

> **Current shipping status (11 November 2026):** `DISTANCE_DELIVERY_ENABLED=false` by default. New checkout uses the existing saved flat shipping configuration, before coupons/points, with free shipping strictly above its saved threshold. Google Routes, dispatch coordinates, slabs and distance quotes are inactive. Historical order charges are unchanged. Distance-specific material below describes a retained, inactive implementation until a future explicitly approved re-enable. See current flat shipping reference.

Retained distance delivery architecture (inactive)

For developers and release owners. Baseline: existing local uncommitted distance-delivery changes, 11 November 2026. Detailed retained implementation/file inventory: DISTANCE_DELIVERY.md. This guide does not establish Google or Production readiness.

Configuration and addresses

Private `siteSettings/dispatchLocation` v1 contains name, full address, latitude, longitude, optional Google Maps URL, active state and revision/audit metadata. `siteSettings/shipping` v2 contains road-distance slabs and free-shipping settings. `siteSettings/codAvailability` v1 controls new COD acceptance. The combined Store Settings service enforces dual Admin authorization, expected revisions and replay protection.

Dispatch is separate from public contact details. A Maps URL is optional metadata; it is not parsed to guess coordinates. Unsaved defaults do not count as approved configuration. A single revision-only shipping signal invalidates customer reviews after origin, pricing or COD changes.

Owned saved addresses keep full postal/contact fields and optional `coordinates: {latitude, longitude}`. Pins do not remove postal requirements. Legacy/manual addresses remain valid inputs. Address changes update revision and invalidate quote identity. Removing a pin deletes its stored coordinate field through the server save path.

Browser map and server routes

The optional picker loads Leaflet code on demand and displays OSM standard tiles with attribution. It supports click/drag and manual coordinates, with manual address entry available if tiles fail. It does not request browser geolocation or perform a Google geocoder call. The initial Hyderabad map centre is not a selected dispatch/customer pin.

`server/routes.ts` calls `https://routes.googleapis.com/directions/v2:computeRoutes` using a server-only key, DRIVE, one primary route, minimal field mask and an eight-second timeout. Selected coordinates are submitted directly; otherwise the complete manual postal address is supplied for provider resolution. Missing/bad routes, partial matches and invalid/provider-error responses fail with a safe generic message. No secret or full customer address is leaked in that error.

Google route data/polylines/resolved positions are not rendered on the OSM map. OSM tile availability, usage suitability and provider terms require operational review. See Google setup.

Quote preparation and validation

1. Existing `orders:review` performs a preliminary trusted product/address/coupon/points review.
2. For a valid address/review, `deliveryQuotes.prepare` reads active origin and road pricing, rate-limits calculation (20 per UID/minute), and calls the route provider outside the order transaction.
3. Fee selection uses the first slab whose upper distance contains the route. The upper bound is inclusive; disabled/out-of-range coverage is unavailable. Free delivery is considered only after coverage succeeds, at or above the threshold after coupons and before points.
4. A private random quote document binds UID, address ID/revision/content identity, destination, origin/pricing revisions, road metres, merchandise basis, fee/free decision and five-minute expiry.
5. Final review and placement transaction re-read quote/settings/address, recompute fee and validate identity, expiry, amounts and revisions. A supplied browser fee cannot authorize an order.

The hook can reuse a matching quote with enough lifetime, refresh before expiry and drop a rejected cached quote before Retry. Quote ID/expiry and quotation time are excluded from the semantic confirmation fingerprint so an identical refresh does not bounce checkout stages. Amounts/settings/address/discount/product terms remain bound; a supplied quote is independently checked.

No quote reserves inventory. COD acceptance and Razorpay reservation consume the final reviewed contract through the existing transaction. Accepted-operation recovery is checked before new settings eligibility. Existing pending online payments can continue reconciliation/resume if new delivery calculation later becomes unavailable.

Failure and historical behaviour

Missing/inactive origin, invalid or legacy-only pricing, absent/invalid key, Google failure, expired/tampered/foreign quote or unavailable range prevents a new delivery checkout. There is no pickup, straight-line fee, production mock switch or flat-fee fallback. Existing historical orders retain their saved totals/addresses and do not require a new route to display.

New order `deliverySnapshot` stores business evidence: quote reference, configuration revisions, slab maximum, fee/free flag and quotation time. Raw Google routes/distance/resolved destination are not copied into that snapshot. Customer-supplied address coordinates are a separate address field and must be covered by the approved privacy/retention policy.

Operational dependencies

- Google Cloud billing, Routes API, restricted server key and quotas: **Pending confirmation**.
- Actual dispatch/slabs/threshold and owner cutover acceptance: **Pending confirmation**.
- TTL/equivalent cleanup for `deliveryQuotes.deleteAfter` and `deliveryQuoteLimits.expiresAt`: **Pending confirmation**. Application expiry does not wait for TTL deletion.
- Real route acceptance and separately approved deployed checkout acceptance: **Pending confirmation**.

Injected route doubles are permitted only with the exact loopback Auth/Firestore emulator ports and no hosted Vercel context. They do not prove real API readiness. No provider response or live fixture should be archived in documentation.

Sources: `server/routes.ts`, `server/deliveryQuotes.ts`, `server/delivery.ts`, `server/orders.ts`, `server/storeSettings.ts`, `src/shipping/distance.ts`, `src/shipping/model.ts`, `src/delivery/LocationPicker.tsx`, `src/delivery/useCheckoutReview.ts`, `src/components/modalLayer.ts`. Local dated checks remain in the retained implementation note; they were not rerun for documentation.